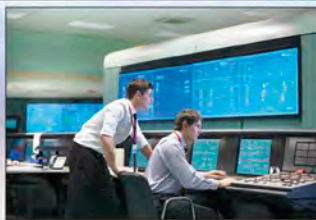




In charge

Power stations can't be built near all the places that need electricity. So the electricity generated flows into a network of cables, called a grid. At the touch of a button, electricity is made to flow to wherever it is needed.



Electricity is dangerous, so tall pylons hold the long cables high above the ground.

Second transformer

Electricity cables are laid under the ground in towns and cities.

A substation makes electricity safe for you to use in your home.

Aluminum cable

Energy travels down the cables at about 155,000 miles (250,000 kilometers) per second—almost as fast as the speed of light!



Going up or going down?

Machines called transformers change the strength of an electric current. The current that flows between pylons has to be decreased to stop the cables from melting.

These children are pretending to be tiny parts of atoms, called electrons.

Each ball is a "parcel" of electrical energy.



Pass the parcel

People once thought that electricity flowed like water, which is why it was called a current. In fact, energy moves along a cable more like balls being passed down a line!

Watt is power?

The speed at which different machines use energy is measured in units called watts.



Electronic clock
(10 watts)



Vacuum cleaner
(900 watts)



Welding machine
(10,000 watts)

